Cybersecurity Network Behaviour Analytics

Project 1 — What Does Normal Network Behaviour Look Like?

Dataset disclosure: The supplied challenge specification defines the required fields and tasks but does not contain actual network-flow rows. This implementation therefore uses a synthetic anonymized dataset for development and demonstration.

1. Objective

Establish normal network behaviour and identify communication patterns that differ strongly from the baseline using traffic volume, duration, connection frequency, protocol, time and source-device activity.

2. Dataset

Fields: Flow_ID, Timestamp, Source, Destination, Bytes_Transferred, Duration, Protocol and Connection_Frequency.

3. Data Preparation

Duplicates were removed, timestamps converted to datetime, numeric fields validated, missing numeric values imputed with medians, and missing protocol values filled with the mode.

4. KNN Method

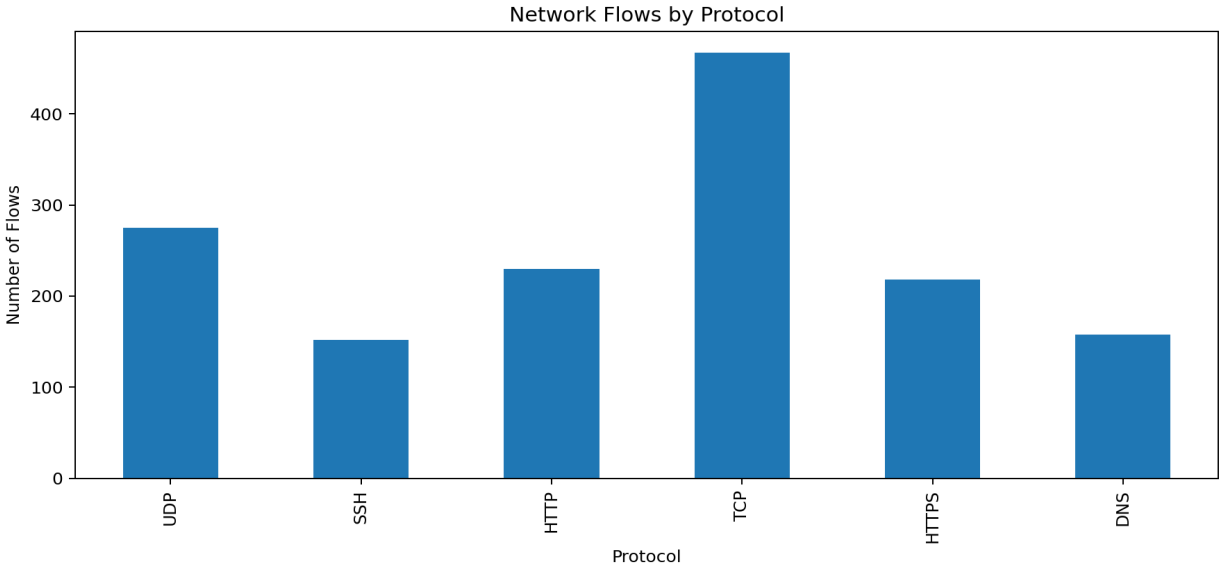
Bytes transferred, duration and connection frequency were standardized. A 10-neighbour KNN model was used. The mean distance to neighbouring records is used as the anomaly score. High scores indicate behaviour that differs from nearby patterns; they do not prove malicious activity.

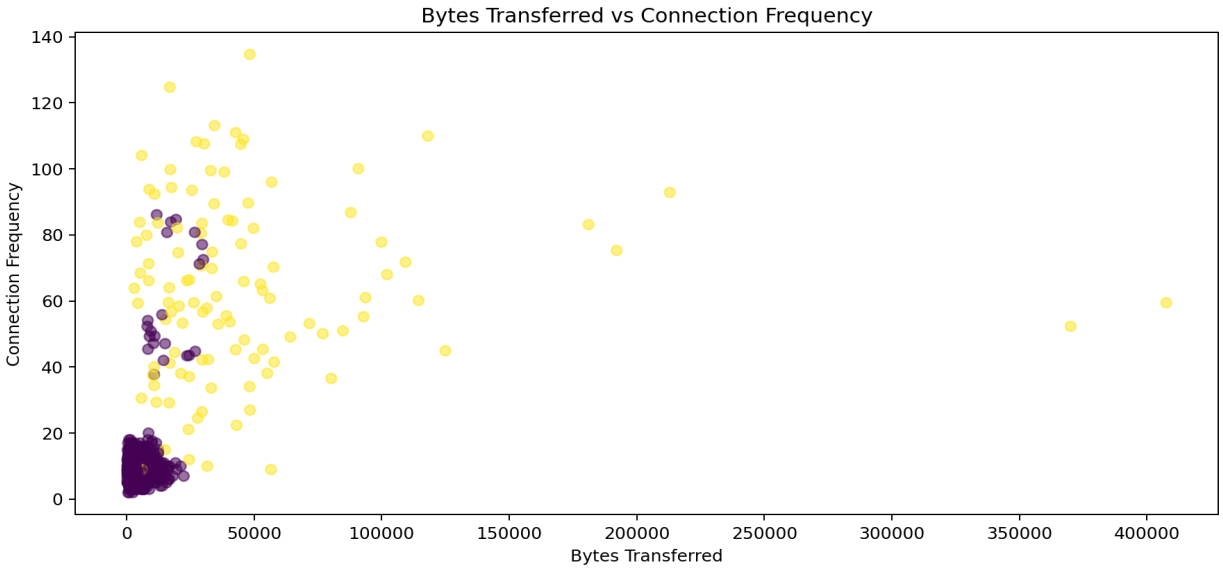
5. Synthetic Validation

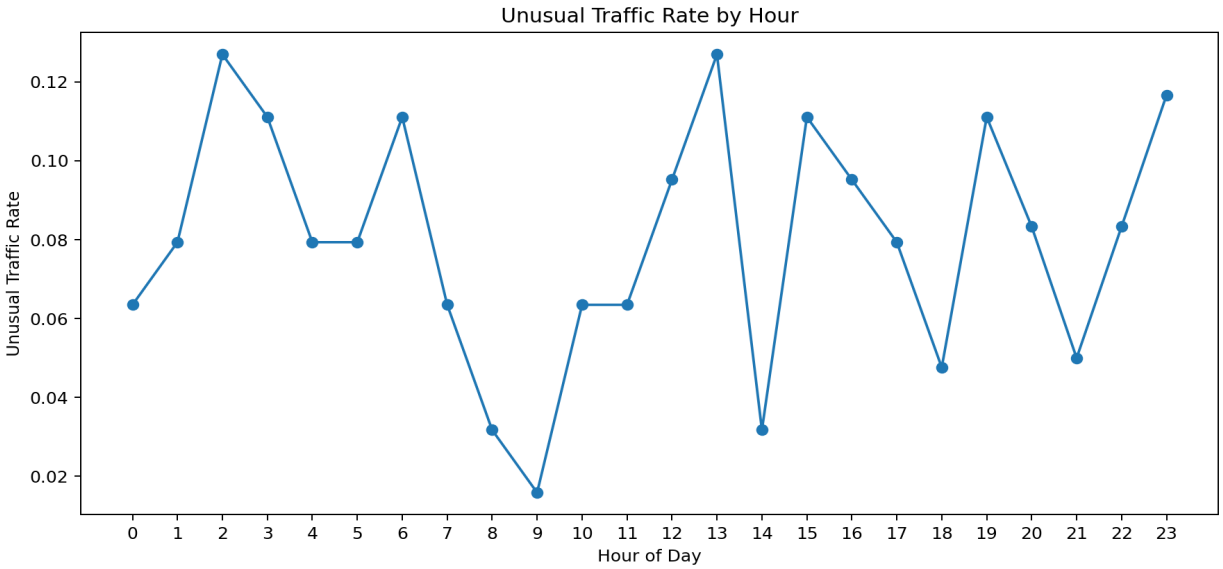
Synthetic validation labels were used only to benchmark the implementation because the supplied challenge contains no attack labels. Precision = 0.867, Recall = 0.812, F1 = 0.839.

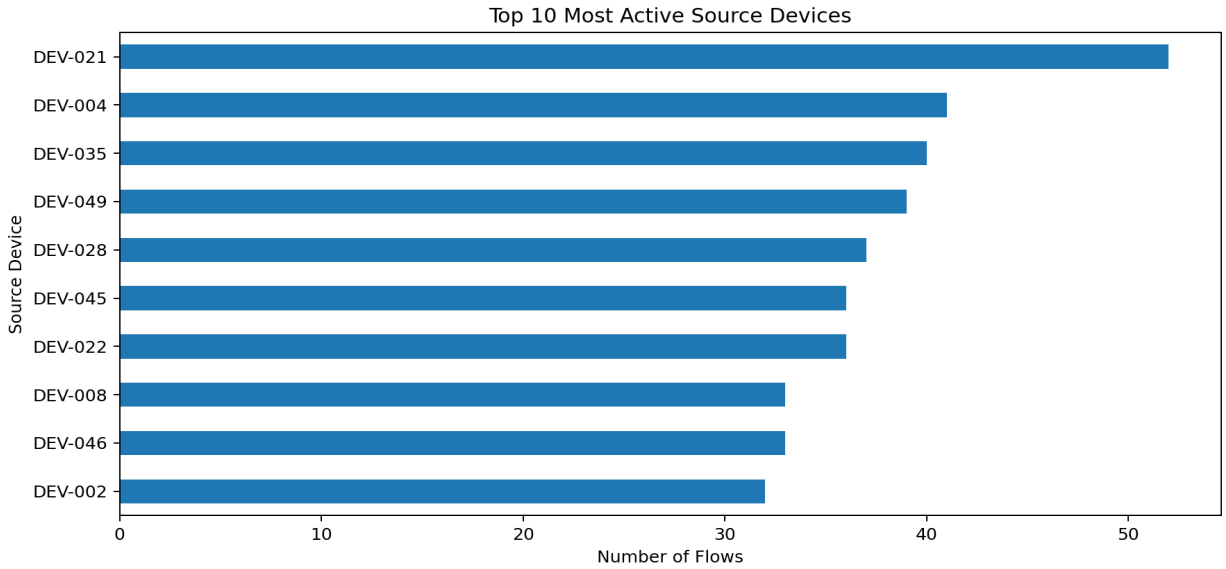
Requirement	Status
Data cleaning and validation	Completed
Exploratory data analysis	Completed
4 visualisations	Completed
Protocol/time/device analysis	Completed
KNN anomaly model	Completed
Evaluation	Completed
5–7 insights and action plan	Completed

6. Required Visualisations









7. Key Insights

1. KNN flags a small high-distance subset as unusual relative to the behavioural baseline.
2. Protocol-level analysis highlights protocol-specific deviations from normal traffic.
3. Source-device activity helps prioritise which devices deserve closer review.
4. High data volume combined with high connection frequency can create strong anomaly signals.
5. Time-of-day analysis provides context for unusual activity outside normal traffic periods.
6. KNN anomaly scores support investigation prioritisation rather than automatic malicious classification.
7. The network baseline should be refreshed periodically as normal behaviour changes.

8. Practical Action Plan

• Prioritise high KNN-score flows.
• Review devices with repeated anomalies.
• Inspect unusually large transfers and connection-frequency spikes.
• Compare anomalies against protocol and hourly baselines.
• Review repeated unusual source-destination pairs.
• Refresh the baseline periodically.
• Require contextual evidence before escalating an anomaly as an incident.

9. Conclusion

The project establishes a network-behaviour baseline, performs EDA, creates four required visualisations, applies KNN-based anomaly detection, evaluates the implementation using synthetic validation labels, and provides a practical investigation workflow.